

CHANDRA BHAKAR

Senior AI Engineer

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CORE EXPERTISE: Generative AI • Multi-Agent Systems • LLMs & RAG • MCP Servers • Vector Databases • Python & Golang • Kubernetes • Enterprise Platform Architecture

PROFESSIONAL SUMMARY

Senior AI Engineer with 8+ years of experience designing and delivering enterprise-scale AI products in Fortune 500 environments. Architected Lowe's flagship AI platform, MyLow Companion, entirely from scratch — before frameworks like LangGraph or Google ADK became mainstream — and scaled it to ~1,900 US stores handling 50,000+ daily queries. Deep expertise spans multi-agent orchestration, MCP server design, RAG pipelines, and high-performance vector retrieval. Consistent record of driving transformational outcomes: 90% cost reduction through AI automation, 500+ internal AI assistants deployed, and sub-second memory retrieval at enterprise concurrency.

IMPACT AT A GLANCE

~1,900 Stores Deployed <i>MyLow Companion across all US locations</i>	50,000+ Daily AI Queries <i>Enterprise-grade availability & reliability</i>	90% Cost Reduction <i>Contact Center GenAI vs. manual vendors</i>	500+ RAG Assistants <i>Across 100+ internal teams at Lowe's</i>
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TECHNICAL SKILLS

AI / ML	Generative AI LLMs Multi-Agent Systems RAG Pipelines FastMCP Agentic Frameworks Prompt Engineering
Languages	Python (Advanced) Golang (Expert) SQL (Expert)
Vector & AI DBs	Milvus Weaviate Google Vertex Vector Search Redis
LLM Tools	OpenAI GPT Mistral VLLM Whisper ASR Google ADK
Infrastructure	Kubernetes Docker Apache Kafka RabbitMQ PostgreSQL
CI/CD & DevOps	Jenkins Argo-CD Kibana FastAPI
Architecture	MCP Servers Microservices System Design TDD Agile

EMPLOYMENT HISTORY

- Senior AI Engineer • **Lowe's India** • Bengaluru Jun 2024 – Present
- Pioneered Agentic Architecture:** Led a team of 4 to design and build MyLow Companion — Lowe's enterprise multi-agent AI platform — from scratch, authoring custom orchestration logic before LangGraph or Google ADK were mainstream.
 - Global Scale Deployment:** Deployed MyLow Companion to ~1,900 US stores, establishing it as the company-wide face of AI with ~50,000 daily associate queries and production-grade availability.
 - Infrastructure Innovation (Memory Service):** Designed a centralized Long-Term Memory MCP server using FastMCP; implemented multi-layered caching (Redis + vector DB), async ingestion, guardrails, and project-level access control — achieving 700–800ms retrieval latency at scale.
 - 90% Cost Reduction:** Designed a GenAI Contact Center pipeline replacing external manual analysis vendors, cutting turnaround from weeks to minutes and delivering a 90% reduction in operational costs.

- **AI Democratization:** Built a self-service Agent/Assistant platform enabling 100+ internal teams (HR, ITSD, Operations) to independently deploy and manage 500+ RAG-based AI assistants.
- **Contact Center Intelligence:** Engineered an end-to-end pipeline transcribing and analyzing 10,000+ call recordings per 8-hour window with automated sentiment scoring and summarization (Whisper + Mistral + VLLM).

Software Engineer • Lowe's India • Bengaluru

Jul 2021 – Jun 2024

- **Automated Data Profiling (ADP):** Built a self-service metadata discovery platform as sole engineer, enabling non-technical users across Lowe's to surface data insights independently.
- **Anomaly Detection Automation (ADA):** Delivered a self-service anomaly detection system with high-speed async worker components for parallel, large-scale data scanning.
- **Automated Continuous Ingestion (ACI):** Developed Golang-based REST schedulers and real-time ingestion pipelines using Kafka and Druid for continuous forecasting workflows.

Software Engineer • Nokia Networks • Bengaluru

Nov 2020 – Jul 2021

- **Network Operation Master (NOM):** Built a cloud-based network management system (FCAPS); led microservices monorepo migration and implemented configuration consistency validation in a TDD/Agile environment.

Associate Software Engineer • TELUS International • Noida

Feb 2018 – Nov 2020

- **Virtual Assistant & Visual IVR:** Designed API payloads and PostgreSQL schemas for enterprise customer service chatbots and digital IVR platforms using Golang and Python.
- **Neustar Analytics:** Developed SOAP APIs for real-time mobile number portability using Oracle SOA, BPEL, and SQL in an Agile delivery environment.

FLAGSHIP PROJECTS

1. MyLow Companion (Lowe's Store Assist AI)

Tech Stack: Python (FastAPI) | OpenAI | Google ADK | Whisper | Mistral | VLLM | Milvus DB | Docker | Kubernetes | Argo-CD

- Architected Lowe's primary enterprise AI product — a multi-agent platform deployed across ~1,900 US stores as the company's public-facing AI identity.
- Enabled voice-to-text and text-to-voice interactions so store associates can resolve product queries, check availability, and compare items hands-free.
- Designed the entire agentic orchestration layer from scratch before LangGraph/ADK existed — demonstrating foundational multi-agent platform expertise.
- Handles ~50,000 daily queries with enterprise-grade availability, observability, and horizontal scalability.

2. Lowe's Memory Service (Enterprise Agent Memory / MCP Server)

Tech Stack: Python | FastMCP | PostgreSQL | Redis | Kubernetes | Weaviate | Google Vertex Vector Search

- Designed a centralized Long-Term Memory service as an MCP server — providing persistent, project-scoped memory for all internal AI agents.
- Implemented multi-layered caching (Redis hot cache + vector DB) with async ingestion pipelines for sub-second memory recall.
- Achieved 700–800ms retrieval latency under enterprise concurrency with project-level access control and security guardrails.
- Serves as reusable AI infrastructure powering memory across MyLow Companion and 500+ RAG assistants.

3. Contact Center Intelligence Initiative

Tech Stack: Python | Whisper ASR | Mistral | VLLM | Milvus DB | Async Worker Architecture

- Built a fully automated end-to-end pipeline transcribing and analyzing 10,000+ call recordings per 8-hour operational window.
- Replaced manual external vendor workflows with AI — achieving 90% cost reduction and cutting turnaround from weeks to minutes.
- Delivered automated sentiment analysis, agent performance scoring, and structured call summaries for CX insights.
- Architected on a scalable async worker model enabling horizontal scale-out during peak processing windows.

EDUCATION

Bachelor of Technology — Computer Science & Engineering

2012 – 2016

JECRC UDML College of Engineering | Rajasthan Technical University, Jaipur